

Maharashtra Public Service Commission Assessment

Subject Name :

Assistant Professor Mathematics Maharashtra Engineering Service
Gr A

Duration :

60

Number of Questions :

100

Total Marks :

200

Display Marks:

Yes

Maximum Instruction Time :

0

Is Section Default? :

No

Question Number : 1 Question Id : 4646621201 Question Type : MCQ

$$\text{Let } f(x, y) = \begin{cases} y \sin \left(\frac{1}{x} \right), & x \neq 0 \\ 0 & , \quad x=0 \end{cases}$$

Which of the following is **correct** ?

Options :

1. $f(x, y)$ is discontinuous at $(0, 0)$
2. $f(x, y)$ is continuous at $(0, 0)$
3. limit of $f(x, y)$ does not exist at $(0, 0)$
4. $f(x, y)$ is differentiable at $(0, 0)$

Question Number : 2 Question Id : 4646621202 Question Type : MCQ

Let $f: \mathbf{R}^2 \rightarrow \mathbf{R}$ be defined by; $f(x,y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$

then the directional derivative of f at $(0,0)$ in the direction of vector $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ is :

Options :

1. $\frac{1}{2}$
2. $\frac{1}{\sqrt{2}}$
3. $\frac{1}{2\sqrt{2}}$
4. $\frac{1}{4\sqrt{2}}$

Question Number : 3 Question Id : 4646621203 Question Type : MCQ

Let C be a positively oriented piecewise smooth Jordan curve bounding a simply connected region D. If $M(x, y)$ and $N(x, y)$ have continuous partial derivatives on D, then Green's theorem states that :

Options :

1.
$$\oint_C (Mdx + Ndy) = \iint_D \left(\frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} \right) dA$$

2.
$$\oint_C (Mdx + Ndy) = \iint_D \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dA$$

3.
$$\oint_C (Mdx + Ndy) = \iint_D \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dA$$

4.
$$\oint_C (Mdx + Ndy) = \iint_D \left(\frac{\partial N}{\partial y} - \frac{\partial M}{\partial x} \right) dA$$

Question Number : 4 Question Id : 4646621204 Question Type : MCQ

For what value of K, the function; $f(x,y) = \begin{cases} 3xy & ; (x,y) \neq (2,3) \\ K & ; (x,y) = (2,3) \end{cases}$ is continuous on $\mathbf{R}^2$?

Options :

1. 6
2. 9
3. 16
4. 18

Question Number : 5 Question Id : 4646621205 Question Type : MCQ

Given that $f(x, y) = 1 - x^2 - y^2$, subject to the constraint $x + y = 1$, with $x \geq 0, y \geq 0$. Find the maximum and minimum values of the function.

Options :

1. Maximum value is 1 and minimum value is 0.
2. Maximum value is 1 and minimum value is $\frac{1}{2}$.
3. Maximum value is $\frac{1}{2}$ and minimum value is $\frac{-1}{2}$.
4. Maximum value is $\frac{1}{2}$ and minimum value is 0.

Question Number : 6 Question Id : 4646621206 Question Type : MCQ

The value of double integral $I = \int_0^4 \int_0^{\sqrt{y}} x \cdot y^2 \, dx \, dy$ is :

Options :

1. 8
2. 12
3. 32
4. 24

Question Number : 7 Question Id : 4646621207 Question Type : MCQ

Find $\oint_C \frac{-y \, dx + x \, dy}{x^2 + y^2}$, where C is any piecewise smooth Jordan curve enclosing the origin (0, 0).

Options :

1. 2π
2. π
3. 4π
4. 0

Question Number : 8 Question Id : 4646621208 Question Type : MCQ

The Taylor's series expansion of function $f(x, y)$ about origin is called _____.

Options :

1. Newton Series
2. Maclaurin Series
3. Fourier Series
4. Laurent Series

Question Number : 9 Question Id : 4646621209 Question Type : MCQ

If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ and $r = |\vec{r}|$ then the value of $\text{grad}(r^n)$ is :

Options :

1. nr^{n-1}
2. $nr^{n-2} \vec{r}$
3. $nr^{n-1} \vec{r}$
4. 0

Question Number : 10 Question Id : 4646621210 Question Type : MCQ

A vector field $\vec{f}$ is called solenoidal if :

Options :

1. $\text{div } \vec{f} = 0$

2. $\text{grad} \cdot \vec{f} = 0$

3. $\text{curl } \vec{f} = \vec{0}$

4. $\nabla \times (\nabla \times \vec{f}) = \vec{0}$

Question Number : 11 Question Id : 4646621211 Question Type : MCQ

$$\text{Let } f(x, y) = \begin{cases} 1, & \text{if } x > 0 \text{ and } y > 0 \\ 0, & \text{otherwise} \end{cases}$$

Which of the following statements is **correct** ?

Options :

1. f is continuous at $(0, 0)$
2. f is differentiable at $(0, 0)$
3. Partial derivatives f_x and f_y do not exist at $(0, 0)$
4. Partial derivatives f_x and f_y exist at $(0, 0)$

Question Number : 12 Question Id : 4646621212 Question Type : MCQ

The operator used to express higher-order derivatives in Taylor's formula for two variables is :

Options :

1. $\frac{\partial}{\partial x} + \frac{\partial}{\partial y}$

2. $\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$

3. $h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y}$

4. $h^2 \frac{\partial^2}{\partial x^2} + hk \frac{\partial^2}{\partial x \partial y} + k^2 \frac{\partial^2}{\partial y^2}$

Question Number : 13 Question Id : 4646621213 Question Type : MCQ

Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be linear transformation such that $T(1, 0) = (1, 2)$ and $T(1, 1) = (0, 2)$ then $T(x, y)$ is equal to :

Options :

1. $(x, 2y)$
2. $(2x, y)$
3. $(x - y, 2x)$
4. $(x - y, 2y)$

Question Number : 14 Question Id : 4646621214 Question Type : MCQ

Given that the set of vectors $\{u, v, w\}$ is linearly independent. Which of the following sets of vectors is also linearly independent ?

Options :

1. $\{u - v, v - w, w - u\}$
2. $\{u + v, u - v, 2u\}$
3. $\{u + v, u - v, u - 2v + w\}$
4. $\{u, v, 0\}$

Question Number : 15 Question Id : 4646621215 Question Type : MCQ

The dimension of the subspace of $\mathbf{R}^3$ spanned by vectors $\bar{u} = (-3, 0, 1)$, $\bar{v} = (1, 2, 1)$ and $\bar{w} = (3, 0, -1)$ is :

Options :

1. 0
2. 3
3. 2
4. 1

Question Number : 16 Question Id : 4646621216 Question Type : MCQ

The value of k for which the system of equations

$$x + y + z = 0$$

$$y + 2z = 0$$

$$kx + z = 0$$

has more than one solution is :

Options :

1. -1

2. 0

3. $\frac{1}{2}$

4. 1

Question Number : 17 Question Id : 4646621217 Question Type : MCQ

The characteristic roots (or eigen values) of the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ are :

Options :

1. 5, 1, 1
2. 5, 2, 2
3. 0, 0, 0
4. 2, 3, 2

Question Number : 18 Question Id : 4646621218 Question Type : MCQ

Suppose a Square matrix A of order n over $\mathbf{R}$ has rank n. Which one of the following statements is **not** correct ?

Options :

1. A^T has rank n
2. A has n-linearly independent columns
3. A is non-singular
4. An eigen value of A is 0

Question Number : 19 Question Id : 4646621219 Question Type : MCQ

Let A be a 5×5 matrix with real entries such that $A^3 = 0$. Then :

Options :

1. The degree of the characteristic polynomial of A is 3.
2. The degree of the characteristic polynomial is 5.
3. Minimal polynomial of A and the characteristic polynomial of A are same.
4. A is invertible.

Question Number : 20 Question Id : 4646621220 Question Type : MCQ

Let $A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}$. Find an orthogonal matrix P such that $D = P^{-1}AP$ is a diagonal matrix.

Options :

1. $P = \frac{1}{\sqrt{10}} \begin{bmatrix} -1 & 3 \\ 3 & 1 \end{bmatrix}$

2. $P = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 & 1 \\ 1 & -3 \end{bmatrix}$

3. $P = \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$

4. $P = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Question Number : 21 Question Id : 4646621221 Question Type : MCQ

If a matrix A with real entries is orthogonal, then :

Options :

1. $\det A = 1$ or -1
2. $A = A^T$
3. A is not invertible
4. $AA^T = 0$

Question Number : 22 Question Id : 4646621222 Question Type : MCQ

Let g be a bilinear form on $\mathbb{R}^2$ defined by; $g((a_1, b_1), (a_2, b_2)) = a_1b_2 + a_2b_1$. Then the matrix of g in the ordered basis $B = \{(1, 1), (1, -1)\}$ of $\mathbb{R}^2$ is :

Options :

1. $\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$

2. $\begin{bmatrix} 0 & 2 \\ 0 & -2 \end{bmatrix}$

3. $\begin{bmatrix} 2 & 0 \\ -2 & -2 \end{bmatrix}$

4. $\begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix}$

Question Number : 23 Question Id : 4646621223 Question Type : MCQ

Let $V(F)$ be a vector space of polynomials in x . An inner product in $V(F)$ is defined as

$\langle p, q \rangle = \int_0^1 p(x) q(x) dx$, where $p = p(x)$, $q = q(x) \in V$. Then find the value of $\langle p, q \rangle$ where

$$p(x) = (x + 2), q(x) = x^2 - 2x - 3.$$

Options :

1. $-\frac{4}{37}$

2. $\frac{4}{37}$

3. $-\frac{37}{4}$

4. $\frac{37}{4}$

Question Number : 24 Question Id : 4646621224 Question Type : MCQ

Let A be a 5×5 matrix and let $B = PAP^{-1}$ for some invertible matrix P . Then :

Options :

1. A and B have same eigen values
2. $A = B$
3. $\det A = \det B$
4. Minimal polynomial of A is same as the characteristic polynomial of B

Question Number : 25 Question Id : 4646621225 Question Type : MCQ

Which of the following best defines the least upper bound of a set S ?

Options :

1. The smallest element of S .
2. The greatest element of S .
3. The smallest number that is greater than or equal to every element of S .
4. The largest number smaller than every element of S .

Question Number : 26 Question Id : 4646621226 Question Type : MCQ

Which one of the following sequence is convergent ?

Options :

1. $\{2^n\}$

2. $\left\{\left(\frac{1}{3}\right)^n\right\}$

3. $\{5^n\}$

4. $\{1 + n^2\}$

Question Number : 27 Question Id : 4646621227 Question Type : MCQ

Let $T : [0, 1] \rightarrow \mathbb{R}$,
 $S : (0, 1) \rightarrow \mathbb{R}$
 $L : [0, 1) \rightarrow \mathbb{R}$
be contraction maps.
Then :

Options :

1. T, S and L are uniformly continuous
2. T is uniformly continuous, and S and L are not continuous
3. T and L are continuous, and S is not continuous
4. T, S and L are not continuous

Question Number : 28 Question Id : 4646621228 Question Type : MCQ

If $f: [a, b] \rightarrow \mathbf{R}$ is a continuous function, then for any $\epsilon > 0$, there exists a polynomial $p(x)$ such that :

Options :

1. $|f(x) - p(x)| < \epsilon$ for all $x \in [a, b]$
2. $|f(x) - p(x)| < \epsilon$ for all $x \in (a, b)$
3. $f(x) = p(x)$ for all $x \in [a, b]$
4. $|f(x) - p(x)| \leq \epsilon$ only at the end points a and b

Question Number : 29 Question Id : 4646621229 Question Type : MCQ

Consider the spaces $\mathbf{Z}$, $\mathbf{Q}$ and $\mathbf{R}$ with the usual metric : $d(a, b) = |a - b|$. Then,

Options :

1. $\mathbf{Z}$ and $\mathbf{Q}$ are complete metric spaces, and $\mathbf{R}$ is not.
2. $\mathbf{Z}$ and $\mathbf{R}$ are complete metric spaces, but $\mathbf{Q}$ is not.
3. $\mathbf{Q}$ and $\mathbf{R}$ are complete metric spaces, but $\mathbf{Z}$ is not.
4. $\mathbf{Z}$, $\mathbf{Q}$ and $\mathbf{R}$ are complete metric spaces.

Question Number : 30 Question Id : 4646621230 Question Type : MCQ

A metric space X is said to be connected if :

Options :

1. Every sequence converges.
2. Every subset of X is closed.
3. X cannot be written as the union of two disjoint non-empty open sets.
4. Every subset of X is bounded.

Question Number : 31 Question Id : 4646621231 Question Type : MCQ

For $f(x, y) = x^2 + 2xy + y^2$, the mixed partial derivative $\frac{\partial^2 f}{\partial x \partial y}$ is :

Options :

1. 1
2. 2
3. $2x$
4. $2y$

Question Number : 32 Question Id : 4646621232 Question Type : MCQ

Let $f_n(x) = \frac{x}{[(n-1)x+1](nx+1)}$ and $S_n(x) = \sum_{j=1}^n f_j(x)$, $x \in [0, 1]$.

Then the sequence $\{S_n\}$ of functions :

Options :

1. Converges uniformly on $[0, 1]$
2. Converges pointwise on $[0, 1]$ but not uniformly
3. Converge pointwise for $x = 0$ but not for $x \in (0, 1]$
4. Does not converge pointwise

Question Number : 33 Question Id : 4646621233 Question Type : MCQ

Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be defined by; $f(x) = \begin{cases} (x-1)^2, & x \text{ rational} \\ 1, & x \text{ irrational} \end{cases}$

Then :

Options :

1. f is continuous everywhere
2. f is continuous at exactly two points
3. f is discontinuous everywhere
4. f is discontinuous at exactly two points.

Question Number : 34 Question Id : 4646621234 Question Type : MCQ

The function; $f(x) = (x - 3)^5 (x - 4)$ has

- I. the point $x = -1$ as a point of minima
- II. the point $x = \frac{7}{9}$ as a point of minima
- III. the point $x = 2$ as a point of maxima

Based on the above statements, choose the **correct** option.

Options :

- 1. All the statements are incorrect.
- 2. II is correct, but I and III are incorrect.
- 3. I and III are correct but II is incorrect.
- 4. III is correct but I and II are incorrect.

Question Number : 35 Question Id : 4646621235 Question Type : MCQ

In the metric space $\mathbf{R}$ with usual metric, consider the sets Z and $[0, 1]$. Then,

Options :

1. Z is closed and compact
2. $[0, 1]$ is compact and Z is open
3. $[0, 1]$ and Z are closed
4. Neither $[0, 1]$ nor Z is compact

Question Number : 36 Question Id : 4646621236 Question Type : MCQ

Consider $f: [0, 1] \rightarrow \mathbf{R}$ defined by; $f(x) = \begin{cases} x \sin \frac{1}{x} & , \quad x \neq 0 \\ 0 & , \quad x = 0 \end{cases}$

Then, at $x = 0$,

Options :

1. f is discontinuous
2. f is continuous but f' is not differentiable
3. f is differentiable but f is not continuous
4. f is continuous and differentiable

Question Number : 37 Question Id : 4646621237 Question Type : MCQ

The limit of a function $u(x, y) = \frac{3xy}{2x^2 + y^2}$ as (x, y) approaches to $(0, 0)$:

Options :

1. is 1
2. is 0
3. does not exist
4. is $\frac{1}{2}$

Question Number : 38 Question Id : 4646621238 Question Type : MCQ

The bilinear transformation $w = T(z)$ that maps the points $z_1 = 0$, $z_2 = i$, and $z_3 = -i$ onto $w_1 = -1$, $w_2 = 1$ and $w_3 = 0$, respectively is :

Options :

1. $\frac{-z - i}{3z - i}$

2. $\frac{-z - i}{3z + i}$

3. $\frac{z + i}{3z - i}$

4. $\frac{z + i}{3z + i}$

Question Number : 39 Question Id : 4646621239 Question Type : MCQ

Let $\phi : \mathbb{C} \rightarrow \mathbb{C}$ be $\phi(z) = e^z$, and

$A = \{z \in \mathbb{C} \mid 0 \leq \operatorname{Re} z \leq 1\}$,

$B = \phi(A)$.

Then B is a :

Options :

1. Closed disc centred at 0
2. Closed annulus centred at 0
3. an open annulus centred at 0
4. an infinite strip with sides parallel to the y -axis

Question Number : 40 Question Id : 4646621240 Question Type : MCQ

If $f(z) = u + i v$ is analytic function, then the total derivative dv is equal to :

Options :

1. $\frac{\partial v}{\partial x} dx - \frac{\partial v}{\partial y} dy$

2. $-\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy$

3. $\frac{\partial u}{\partial x} dx - \frac{\partial u}{\partial y} dy$

4. $\frac{\partial u}{\partial y} dx + \frac{\partial u}{\partial x} dy$

Question Number : 41 Question Id : 4646621241 Question Type : MCQ

The value of line integral $\int_0^{1+i} (x^2 - iy)dz$ along line $y=x$ is :

Options :

1. $-5+i$

2. $\frac{1}{3}(5-i)$

3. $\frac{5-i}{6}$

4. $\frac{3-2i}{6}$

Question Number : 42 Question Id : 4646621242 Question Type : MCQ

If C is a circle $|z| = 2$, then the value of $\int_C \frac{3z^2 + z}{z^2 - 1} dz$ is :

Options :

1. $4\pi i$
2. πi
3. $2\pi i$
4. 0

Question Number : 43 Question Id : 4646621243 Question Type : MCQ

Let γ be a circle in $\mathbb{C}$ of radius 1 with centre 2.5. Then :

$$\frac{1}{2\pi i} \int_{\gamma} \frac{dz}{(z-1)(z-2)(z-3)} =$$

Options :

1. $\frac{-1}{2}$

2. $\frac{1}{2}$

3. $\frac{1}{3}$

4. $\frac{1}{4}$

Question Number : 44 Question Id : 4646621244 Question Type : MCQ

Which of the following maps is nowhere analytic ?

Options :

1. $f(z) = \sin^2 z + \cos^2 z$

2. $f(z) = e^z - 1$

3. $f(z) = |z|^2$

4. $f(z) = z^2 + z + 1$

Question Number : 45 Question Id : 4646621245 Question Type : MCQ

A function $f(x, y)$ is said to satisfy a Lipschitz condition in a region D in xy plane and the points (x, y_1) and (x, y_2) both lie in D if there exists a positive integer M such that :

Options :

1. $|f(x, y_1) - f(x, y_2)| \leq M |y_2 - y_1|$

2. $|f(x, y_1) - f(x, y_2)| \geq M |y_2 - y_1|$

3. $|f(x, y_2) - f(x, y_1)| \leq M |y_2 - y_1|$

4. $|f(x, y_2) - f(x, y_1)| \geq M |y_2 - y_1|$

Question Number : 46 Question Id : 4646621246 Question Type : MCQ

What is the order and degree of differential equation $\frac{d^2y}{dx^2} + \sqrt{1 + \left(\frac{dy}{dx}\right)^5} = 0$?

Options :

1. First order, second degree
2. First order, fifth degree
3. Second order, first degree
4. Second order, second degree

Question Number : 47 Question Id : 4646621247 Question Type : MCQ

Solution of equation $\frac{d^4y}{dx^4} + 8 \frac{d^2y}{dx^2} + 16 = 0$ is :

Options :

1. $y = (c_1 + c_2x) \cos 2x + (c_3 + c_4x) \sin 2x$
2. $y = e^x [(c_1 + c_2x) \cos 2x + (c_3 + c_4x) \sin 2x]$
3. $y = e^x [(c_1 + c_2x) \sin 2x + (c_3 + c_4x) \cos 2x]$
4. $y = (c_1 + c_2x) \sin 2x + (c_3 + c_4x) \cos 2x$

Question Number : 48 Question Id : 4646621248 Question Type : MCQ

The solution of Cauchy's Euler differential equation

$$x^3 y''' + xy' - y = 3x^4 \text{ is :}$$

Options :

1. $y = c_1x + c_2x^2 \log x + c_3x \log^2 x + \frac{1}{9}x^4$

2. $y = c_1x^2 + c_2x \log x + c_3x \log^2 x + \frac{1}{9}x^4$

3. $y = c_1x + c_2x \log^2 x + c_3x \log^3 x + \frac{1}{9}x^4$

4. $y = c_1x + c_2x \log x + c_3x \log^2 x + \frac{1}{9}x^4$

Question Number : 49 Question Id : 4646621249 Question Type : MCQ

Let $f(x, y)$ and $\frac{\partial f}{\partial y}$ be continuous functions of x and y on a closed rectangle R with sides parallel to the axes. If (x_0, y_0) is any interior point of R , then there exists a number $h > 0$ with the property that the initial value problem $\frac{dy}{dx} = f(x, y), y(x_0) = y_0$ has :

Options :

1. Infinitely many solutions
2. Two solutions on the interval $|x - x_0| \leq h$
3. One and only one solution $y = y(x)$ on the interval $|x - x_0| \leq h$
4. Has no solution

Question Number : 50 Question Id : 4646621250 Question Type : MCQ

Solving the differential equation $y'' - 2y' + y = e^x \cdot \log x$ by variation of parameter, the value of Wronskian W is :

Options :

1. 2
2. e^{2x}
3. e^{-2x}
4. e^{4x}

Question Number : 51 Question Id : 4646621251 Question Type : MCQ

The general solution of equation

$$y'' - y' - 2y = 4x^2 \text{ is :}$$

Options :

1. $y = c_1 e^{2x} + c_2 e^{-x} - 3 + 2x - 2x^2$
2. $y = c_1 e^{-2x} + c_2 e^{-x} + 4$
3. $y = c_1 e^{2x} + c_2 e^{-x} + 3 - 2x + 2x^2$
4. $y = c_1 e^{-2x} + c_2 e^x - 3 - 2x + 2x^2$

Question Number : 52 Question Id : 4646621252 Question Type : MCQ

To solve a Cauchy-Euler equation, we assume a solution of the form :

Options :

1. $y = e^{mx}$
2. $y = x^m$
3. $y = \sin(mx)$
4. $y = \ln(x)$

Question Number : 53 Question Id : 4646621253 Question Type : MCQ

Bessel function $J_n(x)$ all solutions of which differential equation :

Options :

1. $x^2 y'' + xy' + (x^2 - n^2)y = 0$
2. $x^2 y'' - xy' + (x^2 - n^2)y = 0$
3. $xy'' + y' + xy = 0$
4. $xy'' + n^2y = 0$

Question Number : 54 Question Id : 4646621254 Question Type : MCQ

Three linearly independent solutions to differential equation

$$y''' - 4y'' + 5y' - 2y = 0$$

are given by :

Options :

1. $\{e^x, e^{2x}, xe^{2x}\}$
2. $\{e^x, e^{2x}, e^{3x}\}$
3. $\{e^x, xe^x, x^2 e^x\}$
4. $\{e^x, xe^x, e^{2x}\}$

Question Number : 55 Question Id : 4646621255 Question Type : MCQ

Let Z is the set of integers, also let $R = Z + iZ = \{a + ib; a, b \in Z\}$. Then which are the associates in R ?

Options :

1. $a = 1 - i\sqrt{2}; b = \sqrt{2} - i$
2. $a = 1 + i\sqrt{2}; b = \sqrt{2} - i$
3. $a = 2 + i\sqrt{2}; b = \sqrt{2} + i$
4. $a = 1 + i\sqrt{2}; b = 1 - i\sqrt{2}$

Question Number : 56 Question Id : 4646621256 Question Type : MCQ

Let F is the field of rational numbers and α is an element of an extension field K of F . Also assume that $\alpha = \sqrt{2} + \sqrt{3}$ is algebraic over F . Then the irreducible polynomial for α over F is :

Options :

1. $x^4 + 10x^2 + 1$
2. $x^4 - 10x^2 + 1$
3. $x^4 + 10x^2 - 1$
4. $x^4 + 20x^2 + 1$

Question Number : 57 Question Id : 4646621257 Question Type : MCQ

A group G of order 42 has a unique :

Options :

1. Sylow 3 - subgroup
2. Sylow 4 - subgroup
3. Sylow 7 - subgroup
4. Sylow 6 - subgroup

Question Number : 58 Question Id : 4646621258 Question Type : MCQ

A group of order 31 is :

Options :

1. Symmetric Group
2. Cyclic Group
3. Quotient Group
4. Product Group

Question Number : 59 Question Id : 4646621259 Question Type : MCQ

If H and K are finite subgroups of G such that $G = HK$ and $H \cap K = \{e\}$ then :

Options :

1. $O(G) = O(H) O(K)$
2. $O(G) = O(H)$
3. $O(G) = O(K)$
4. $O(G) = O(H \cap K)$

Question Number : 60 Question Id : 4646621260 Question Type : MCQ

Let p and q be integers different from zero and r be their least common multiple. Also r is the positive integer that generates the subgroup $S = \mathbb{Z}_p \cap \mathbb{Z}_q$. So $\mathbb{Z}_r = \mathbb{Z}_p \cap \mathbb{Z}_q$. Then r is divisible by :

Options :

1. Both p and q
2. Only by p
3. Only by q
4. Not divisible by both p and q

Question Number : 61 Question Id : 4646621261 Question Type : MCQ

Let $\phi : G \rightarrow G'$ be a homomorphism of groups and let K be the kernel of ϕ . Then K is :

Options :

1. Quotient Group
2. Normal Subgroup
3. Cyclic Subgroup
4. Symmetric Group

Question Number : 62 Question Id : 4646621262 Question Type : MCQ

Let G be finite group of order n , and let p be prime dividing n . Which of the following is always true ?

Options :

1. If $p^2 \mid n$ then G has an element of order p^2 .
2. G has subgroup of order p^m for every m such that $p^m \mid n$.
3. Every subgroup of order p^k ($p^k \mid n$) is normal in G .
4. If p^k is the highest power of p dividing n , then G has unique subgroup of order p^k .

Question Number : 63 Question Id : 4646621263 Question Type : MCQ

Let $T : D(T) \rightarrow Y$ be a bounded linear operator with domain $D(T) \subset X$, where X and Y are normed spaces. Then :

(I) If $D(T)$ is closed subset of X , then T is closed.

(II) If T is closed, then $D(T)$ is a closed subset of X .

Then choose the **correct** option :

Options :

1. Statement (I) is incorrect but (II) is correct.
2. Both (I) and (II) are correct statements.
3. Statement (II) is incorrect but (I) is correct.
4. Both (I) and (II) are incorrect statements.

Question Number : 64 Question Id : 4646621264 Question Type : MCQ

Let V be a finite dimensional inner product space. Then V has an :

Options :

1. Orthogonal set
2. Self orthogonal
3. Orthogonal basis
4. Orthonormal basis

Question Number : 65 Question Id : 4646621265 Question Type : MCQ

Let Y and Z be subspaces of a normed space X are suppose that Y is closed and is a proper subset of Z . Then for every real number θ in the interval $(0, 1)$ there is a $z \in Z$ such that :

Options :

1. $\|z\| = 0 ; \|z-y\| \leq \theta \quad \forall y \in Y$
2. $\|z\| = 1 ; \|z-y\| \leq \theta \quad \forall y \in Y$
3. $\|z\| = 0 ; \|z-y\| \geq \theta \quad \forall y \in Y$
4. $\|z\| = 1 ; \|z-y\| \geq \theta \quad \forall y \in Y$

Question Number : 66 Question Id : 4646621266 Question Type : MCQ

A set $S = \{\bar{e}_1, \bar{e}_2, \dots, \bar{e}_n\}$ in an Hilbert space is said to be orthonormal if _____.

Options :

1. $\|\bar{e}_n\| = 0$
2. $\|\bar{e}_i\| = 1$ and $\langle e_i, e_j \rangle = 0$ for $i \neq j$
3. All vectors are orthogonal only
4. All vectors have norm 2

Question Number : 67 Question Id : 4646621267 Question Type : MCQ

A closed graph theorem states that a linear operator between Banach spaces is bounded if _____.

Options :

1. its kernel is closed
2. its range is closed
3. its graph is closed
4. it is surjective

Question Number : 68 Question Id : 4646621268 Question Type : MCQ

A metric $d(x, y)$ induced by a norm on space X satisfies the additional conditions :

Options :

1. $d(x+z, y+z) = d(x, y) \forall x, y \in X$
2. $d(\alpha x, \alpha y) = |\alpha| d(x, y) \forall \text{ scalar } \alpha$
3. Both $d(x+z, y+z) = d(x, y) \forall x, y \in X$ and $d(\alpha x, \alpha y) = |\alpha| d(x, y) \forall \text{ scalar } \alpha$ holds
4. None of above is correct

Question Number : 69 Question Id : 4646621269 Question Type : MCQ

From the following table, find p where p is any real number $[\because x = x_0 + ph]$ by Newton's forward interpolation formula, the number of students who obtained marks between 40 and 45.

Marks	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
No. of Students	31	42	51	35	31

Options :

1. $p = 0.2$
2. $p = 0.3$
3. $p = 0.4$
4. $p = 0.5$

Question Number : 70 Question Id : 4646621270 Question Type : MCQ

A solution of the First Approximation of the ordinary differential equation $\frac{dy}{dx} = y + x$ where $y = 1$ when $x = 0$ is given by :

Options :

1. $y_1 = \frac{1}{2}x^2 - x - 1$

2. $y_1 = -\frac{1}{2}x^2 - x - 1$

3. $y_1 = -\frac{1}{2}x^2 + x - 1$

4. $y_1 = \frac{1}{2}x^2 + x + 1$

Question Number : 71 Question Id : 4646621271 Question Type : MCQ

The fixed point of $g(x) = \frac{2x+3}{x+2}$ is :

Options :

1. $\frac{-3}{2}$ and $\frac{3}{2}$

2. $\frac{-2}{3}$ and $\frac{2}{3}$

3. $\sqrt{3}$ and $-\sqrt{3}$

4. $\sqrt{2}$ and $-\sqrt{2}$

Question Number : 72 Question Id : 4646621272 Question Type : MCQ

The error in Trapezoidal rule for $\int_{x_0}^{x_1} y dx$ with n subintervals of width h is, where $y''(\bar{x})$ is the largest value :

Options :

1. $\frac{-(b-a)h^2}{12} y''(\bar{x})$

2. $\frac{-(b-a)h^4}{180} y''(\bar{x})$

3. $\frac{-(b-a)h}{2} y''(\bar{x})$

4. $\frac{-(b-a)h^3}{12} y''(\bar{x})$

Question Number : 73 Question Id : 4646621273 Question Type : MCQ

The spectral radius of matrix A is where $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ -1 & 1 & 1 \end{bmatrix}$

Options :

1. 1
2. 2
3. $\sqrt{3}$
4. 0

Question Number : 74 Question Id : 4646621274 Question Type : MCQ

The Newton-Raphson method fails when :

Options :

1. $f'(x)$ is zero
2. $f'(x)$ is negative
3. $f'(x)$ is too large
4. $f'(x)$ is positive

Question Number : 75 Question Id : 4646621275 Question Type : MCQ

In case of Bisection method, the convergence is :

Options :

1. Linear
2. Quadratic
3. Cubic
4. Very slow

Question Number : 76 Question Id : 4646621276 Question Type : MCQ

The number of strips required in Simpson's $\frac{3}{8}$ th rule is a multiple of :

Options :

1. 1
2. 2
3. 3
4. 6

Question Number : 77 Question Id : 4646621277 Question Type : MCQ

If $z = x \cos xy$ then the value of $\frac{\partial z}{\partial x}$ is :

Options :

1. $xy \sin xy + \cos xy$
2. $-xy \sin xy + \cos xy$
3. $-xy \sin xy - \cos xy$
4. $xy \sin xy - \cos xy$

Question Number : 78 Question Id : 4646621278 Question Type : MCQ

General solution of $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ is of the form _____.

Options :

1. $u = f(x + iy) + g(x - iy)$
2. $u = f(x - iy) + g(x + iy)$
3. $u = f(x + iy) - g(x - iy)$
4. $u = f(x - iy) - g(x - iy)$

Question Number : 79 Question Id : 4646621279 Question Type : MCQ

The Laplace equation is a second order partial differential equation. Which is of :

Options :

1. elliptic type
2. parabolic type
3. hyperbolic type
4. neumann type

Question Number : 80 Question Id : 4646621280 Question Type : MCQ

What is the Canonical form of a hyperbolic equation ?

Options :

1. $\frac{\partial^2 z}{\partial \xi \partial \eta} = \phi(\xi, \eta, z, z_\xi, z_\eta)$

2. $\frac{\partial^2 z}{\partial \xi^2} = \phi(\xi, \eta, z, z_\xi, z_\eta)$

3. $\frac{\partial^2 z}{\partial \xi^2} + \frac{\partial^2 z}{\partial \eta^2} = \phi(\xi, \eta, z, z_\xi, z_\eta)$

4. $\frac{\partial z}{\partial \xi} = \phi(\xi, \eta, z_\xi, z_\eta)$

Question Number : 81 Question Id : 4646621281 Question Type : MCQ

When applying the method of separation of variables to the two dimensional Laplace equation, which form of the solution is assumed ?

Options :

1. $u(x, y) = X(x) + Y(y)$
2. $u(x, y) = X(x)Y(y)$
3. $u(x, y) = e^{X(x)Y(y)}$
4. $u(x, y) = X(x) / Y(y)$

Question Number : 82 Question Id : 4646621282 Question Type : MCQ

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be any twice continuously differentiable function, and $g : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be $g(x, t) = f(x - t)$; then g is a solution of the :

Options :

1. Heat equation
2. Laplace equation
3. Hermite equation
4. Wave equation

Question Number : 83 Question Id : 4646621283 Question Type : MCQ

A solution containing arbitrary constants equal in number to the order of PDE is called :

Options :

1. Particular solution
2. Singular solution
3. Complete integral
4. General solution

Question Number : 84 Question Id : 4646621284 Question Type : MCQ

The solution of partial differential equation $yq - xp = z$ is :

Options :

1. $F\left(\frac{y}{x}, \frac{y}{z}\right) = 0$

2. $F\left(\frac{x}{y}, \frac{x}{z}\right) = 0$

3. $F\left(\frac{z}{x}, \frac{z}{y}\right) = 0$

4. $F\left(xy, \frac{y}{z}\right) = 0$

Question Number : 85 Question Id : 4646621285 Question Type : MCQ

If $X = \{a, b\}$, $P(X) = \{\phi, X, \{a\}, \{b\}\}$ then which one of the following is a topology ?

Options :

1. $\tau = \{\phi, \{a\}, \{b\}\}$
2. $\tau = \{X, \{a\}, \{b\}\}$
3. $\tau = \{\{a\}, \{b\}\}$
4. $\tau = \{\phi, X\}$

Question Number : 86 Question Id : 4646621286 Question Type : MCQ

If f is continuous function from a topological space X into a topological space Y , then :

Options :

1. $f^{-1}(A)$ is closed in X if A is closed subset of Y .
2. $f^{-1}(A)$ is compact if A is compact subset of Y .
3. $\overline{f(A)} \subset f(\overline{A})$, for every subset A of X .
4. None of the above

Question Number : 87 Question Id : 4646621287 Question Type : MCQ

Which of the following subset is connected ?

Options :

1. $\{(x, y) \mid |x^2 - y^2| \geq 4\}$ of $\mathbb{R}^2$
2. $A = \{a, c, e\}$ subset of a topological space (X, τ) , where $X = \{a, b, c, d, e\}$
3. Real line $\mathbb{R}$, with usual topology
4. $A = \{x \in \mathbb{R} \mid |x| > 2\}$ as a subset of real line with usual topology

Question Number : 88 Question Id : 4646621288 Question Type : MCQ

If B is the collection of all intervals in the real line $(a, b) = \{x \mid a < x < b\}$, the topology generated by B is called the :

Options :

1. standard topology on the real line.
2. discrete topology on the real line.
3. hausdorff topology on the real line.
4. product topology on the real line.

Question Number : 89 Question Id : 4646621289 Question Type : MCQ

Let A be a connected subset of topological space X . If $A \subset B \subset \overline{A}$, then B is _____ subset of X .

Options :

1. separable
2. connected
3. disconnected
4. none of these

Question Number : 90 Question Id : 4646621290 Question Type : MCQ

Let (X, d) be a metric space and let $\{x_n\} \subset X$ satisfy

$d(x_n, x) \rightarrow 0$ and $d(x_n, y) \rightarrow 0$. Which of the following is necessarily true ?

Options :

1. $x \neq y$
2. $d(x, y) = 1$
3. $x = y$
4. The sequence $\{x_n\}$ is eventually constant

Question Number : 91 Question Id : 4646621291 Question Type : MCQ

Let X be a normal topological space and let A and B be two disjoint closed subsets of X . According to Urysohn's Lemma there exists a continuous function $f: X \rightarrow [0, 1]$ such that :

Options :

1. $f(A) = 0, f(B) = 1$
2. $f(A) = 1, f(B) = 0$
3. Either A or B
4. $f(A) = f(B) = 1/2$

Question Number : 92 Question Id : 4646621292 Question Type : MCQ

Let $q: X \rightarrow Y$ be a quotient map. Which of the following statements is always true ?

Options :

1. Every continuous surjective map is a quotient map
2. Every quotient map is continuous and surjective
3. Every quotient map is an open map
4. Every quotient map is a closed map

Question Number : 93 Question Id : 4646621293 Question Type : MCQ

Which method is used to find optimum solution of assignment problem ?

Options :

1. Least cost method
2. Hungarian method
3. North-west corner method
4. Vogel's Approximation Method (VAM)

Question Number : 94 Question Id : 4646621294 Question Type : MCQ

To find dual of linear programming problem, the primal must be in _____.

Options :

1. Standard form
2. Canonical form
3. With $\leq, =, \geq$ signs
4. Orthogonal form

Question Number : 95 Question Id : 4646621295 Question Type : MCQ

A set of feasible solutions in linear programming problem is _____.

Options :

1. Non-convex set
2. Concave set
3. Convex set
4. Disconnected set

Question Number : 96 Question Id : 4646621296 Question Type : MCQ

In linear programming problem _____.

Options :

1. objective function, constraints and variables are all linear.
2. only objective function to be linear.
3. only constraints are to be linear.
4. only variables are to be linear.

Question Number : 97 Question Id : 4646621297 Question Type : MCQ

A variable used to convert a greater than or equal to ($\geq$) constraint into equality by subtracting it from left side of the constraints in simplex method is known as :

Options :

1. Slack Variable
2. Artificial Variable
3. Surplus Variable
4. Redundant Variable

Question Number : 98 Question Id : 4646621298 Question Type : MCQ

In cutting plane algorithm each cut involves the introduction of :

Options :

1. Less than or equal to constraint
2. An equality constraint
3. Greater than or equal to constraint
4. An Artificial Variable

Question Number : 99 Question Id : 4646621299 Question Type : MCQ

In Mixed Integer Programming Problem :

Options :

1. Different objective functions are mixed together.
2. All of the decision variables require integer solution.
3. Only few of the decision variables require integer solutions.
4. All constraints functions are mixed together.

Question Number : 100 Question Id : 4646621300 Question Type : MCQ

Given $\sum a_{ij} x_j \leq b_i$, to convert this constraint into equality we introduce _____.

Options :

1. surplus variable
2. unrestricted variable
3. a constant
4. slack variable